

User Churn Project | ML Model Results

Prepared for: Waze Leadership Team

➤ ISSUE / PROBLEM

Waze is losing active app users (churn), which reduces engagement and revenue.

The business lacked a system to identify which users were likely to leave before they stopped using the app.

The dataset suffers from severe **class imbalance** (far fewer churned users than retained users), making it difficult for models to accurately detect churn without predicting false positives.

➤ IMPACT

Improved Churn Detection: Almost doubled the ability to identify churn-prone users compared to the prior baseline, creating a reliable foundation for proactive retention.

Targeted Retention Campaigns: Enabled marketing and product teams to deploy low-risk, proactive interventions, such as customized traffic alerts, driving summary insights, or targeted push notifications, directly to high-risk drivers.

Next Action : Threshold Optimization: Adjust the classification decision threshold to favor higher recall, capturing an even larger pool of at-risk drivers since the business cost of a false positive is minimal.

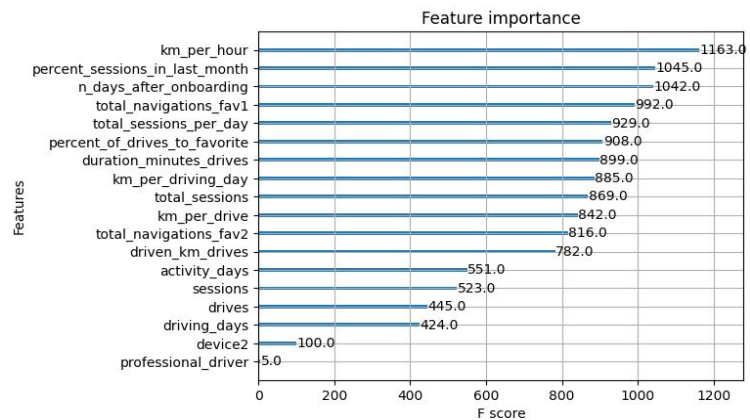
Next Action : Production Monitoring & Iteration: Pilot the XGBoost model on a live user cohort to track campaign conversion rates, while gathering additional granular activity data to engineer new features for future model iterations.

➤ RESPONSE

Engineered tree-based machine learning models (Random Forest and XGBoost) using engineered behavior metrics, such as kilometers driven per day, percentage of monthly sessions, and trip ratios, to predict driver churn.

Evaluated the models using a recall-focused strategy, selecting **XGBoost** as the champion model after it achieved a ~17.15% recall, nearly doubling baseline performance while maintaining an 81.09% overall accuracy.

➤ KEY INSIGHTS



Driving Density Drives Retention: **km_per_driving_day** and **driven_km_drives** are top predictors; users who drive longer distances per session are far more likely to stay engaged.

User Tenure & Activity: Account age (**n_days_after_onboarding**) combined with recent app activity (**percent_sessions_in_last_month** and **total_sessions**) clearly separates retained users from high-risk churners.

App Time vs. Distance: Total navigation time (**duration_minutes**) serves as a key signal; a sharp drop in session duration usually signals impending churn.

Low-Value Features: Static user metadata (such as device type) provides almost no predictive value and can be deprioritized in future model iterations.

Targeted Interventions: Use drop-offs in recent session activity and daily driving distance as automated triggers for targeted re-engagement offers before users abandon the app completely.